



Ioannis BEKAS

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English (fluent), Greek (native)

>> EDUCATION

Technical University of Crete & Hellenic Army Academy | M.Sc. Operational Research, Analytics & Decision Making

GPA: 9.3 / 10.0 Thesis: "[Artificial Intelligence Touchpoints with Multi-Criteria Decision Analysis](#)"

University of Athens | B.Sc. Mathematics — Minor in Economics

Exchange semester: Financial Mathematics — Stockholm University

- **Data Science Professional Certificate** — HarvardX
- **Google Data Analytics Professional Certificate** — Google
- **Financial Engineering and Risk Management** — Columbia University
- **Financial Markets** — Yale University
- **Python and Statistics for Financial Analysis** — Hong Kong University of Science and Technology (HKUST)

Full credentials available at: [linkedin.com/in/ioannis-bekas](https://www.linkedin.com/in/ioannis-bekas)

>> PROFESSIONAL EXPERIENCE

University of Zurich UZH – Department of Finance | AI Engineer & Research Associate (contract)

Apr 2026 – Aug 2026 | Zurich, Switzerland (Remote) – 60% part time

- Architected and implemented the Climate Contribution Evidence Engine (CC-EE), a reproducible AI research pipeline for parsing corporate sustainability reports, annual reports, and SEC 10-K filings into structured, source-linked evidence units. Scaled the workflow to a STOXX 600 latest-report sample, producing 5,880 diagnostic evidence rows across 392 scored firms and 15 climate-transition indicators.
- Designed the model-evaluation workflow for climate-disclosure evidence assessment, combining keyword retrieval, semantic recall audits, LLM-as-judge diagnostics, frozen ClimateBERT classifiers, test-retest checks, OpenAI API reproducibility runs, and AI-polishing robustness experiments. Maintained strict separation between candidate evidence, AI-assisted diagnostic labels, and future human-validated research variables.
- Built research-grade data and validation infrastructure, including schemas, firm-identifier mapping, market-data readiness checks, report-date gates, CRSP/STOXX linkage audits, run manifests, fail-closed preflight scripts, and reproducibility dashboards to prevent exploratory outputs from being promoted into unsupported empirical claims.
- Delivered manager- and paper-ready artifacts, including Colab demos, Overleaf tables, Excel trackers, PowerPoint briefings, runbooks, and validation notes communicating pipeline design, reliability evidence, empirical limitations, and next-step data requirements to academic supervisors and project stakeholders.

United Nations (2 years 8 months)

Nov 2023 – June 2026 | Remote

UN Women | Emergency Data Analyst

- Reviewed Gaza emergency site-sweep XLSForm materials, humanitarian stakeholder correspondence, public Power BI dashboard outputs, and data-sharing requirements; mapped gender-responsive indicators, priority request fields, public-data availability, protection-sensitive schema risks, and reliability caveats for crisis-response analysis.
- Built and maintained humanitarian analytics dashboards using Python, JavaScript, Chart.js, MapLibre, HTML/CSS, and automated validation scripts, including processed data layers, multilingual UI support, comparison tables, map interactions, methodology/source panels, responsive fixes, and visual QA.
- Developed reproducible food-security, displacement, and shock-exposure data pipelines integrating WFP/HDX prices, IPC, WDI, FAOSTAT, UNHCR, OCHA FTS, ILOSTAT, IDMC/IOM displacement, Findex, and related public evidence layers to support prioritisation for crisis-affected populations.
- Designed and documented a Shock Exposure Index workflow for humanitarian prioritisation, covering component scoring, weighting, missing-data treatment, before/after food-price screening, gender/labour vulnerability indicators, dashboard-only workbooks, and interpretation rules separating prioritisation evidence from causal claims.
- Produced handover-ready Excel, CSV, JSON, Word, PDF, PowerPoint, and static-web artifacts for humanitarian colleagues and partners, including source inventories, data dictionaries, validation reports, methodology notes, colleague packages, and deployment copies.
- Strengthened humanitarian data governance through source-of-truth documentation, output manifests, retention rules, dry-run audit scripts, workspace health reports, cleanup approval logs, and a local project cockpit to support reproducibility, accountability, and programme continuity.

United Nations Office for Disaster Risk Reduction (UNDRR) | Data Science & AI Consultant

- Developed AI-driven Early Warning System (EWS) models in Python using scikit-learn, TensorFlow/PyTorch, XGBoost, feature engineering, transfer learning, and time series forecasting to predict tropical storms, droughts, and floods and support anticipatory action, preparedness, and risk-informed humanitarian planning; work awarded internationally and presented at GP2025, a UN General Assembly-mandated forum in Geneva.
- Built and maintained end-to-end ETL/ELT pipelines in Python, SQL, Spark, Airflow-style workflows, and API integrations ingesting from GRI, EM-DAT, World Bank, and OECD CRS to feed predictive models and a live Power BI Global EWS Investment Observatory for disaster risk financing and preparedness decision-making.
- Engineered SQL databases, PostgreSQL data marts, star-schema data models (DAX), and Snowflake/BigQuery/Redshift-aligned warehouse patterns optimized for scalable humanitarian and disaster-risk analytics; established global data governance taxonomy across 150+ countries.
- Integrated climate, population, and accessibility datasets as model features to enrich situational awareness for disaster-prone and hard-to-reach communities and support UN policy decision-makers.
- Delivered quantitative analyses on financial flows and preparedness funding gaps, translating ML outputs into policy-actionable insights for senior officials working on disaster risk reduction and humanitarian resilience.

International Organization of Migration (IOM) | Data Analysis and Web Analytics Consultant

- Collected and structured migration and displacement data from organizational databases, surveys, and field reports using data integration tools and cloud-based storage solutions to support evidence-informed programme delivery.
- Tracked and assessed web traffic on UN migration-related websites through Google Analytics to understand information needs, user engagement, and content effectiveness for migrant and crisis-affected audiences.
- Applied statistical techniques, hypothesis testing, time series forecasting, simulation, and machine learning in Python/R to interpret migration data, identify operational trends, and provide actionable insights for humanitarian and development programming.
- Created reports, simulations, and visual representations that conveyed complex migration findings to non-technical stakeholders, including policy makers and programme managers, using Power BI Service.
- Collaborated across policy, communications, operations, and IT teams to promote data-driven decision-making in migration programmes and field-informed strategies.
- Automated workflows from Qualtrics to Power BI, establishing APIs and visualizing survey findings to accelerate feedback loops for programme monitoring and stakeholder reporting.
- Ensured data accuracy and reliability through routine quality checks and data validation processes for humanitarian reporting and operational decision support.
- Stayed current with developments in data analysis, web analytics, and migration research to strengthen operational analytical approaches for vulnerable and mobile populations.
- Offered data-driven insights to support the development, monitoring, and evaluation of organizational policies and programmes serving migrants and affected communities.
- Engaged stakeholders, including UN officials, through stakeholder management, cross-functional collaboration, mentoring, and training in data analysis and web analytics tools, while guiding teams in using cloud-based platforms for informed operational and programme decision-making.

BI Solutions Group — Advanced Analytics & AI Consulting | (2 years 4 months)

Aug 2021 – Nov 2023 | Remote

- Deployed production ML pipelines (classification, recommendation engines, graph analytics) in Databricks/Spark Feature Store with Git-based CI/CD automation, Docker, Kubernetes-ready containers, GitHub Actions, unit testing, and pytest-style validation for Fortune 500 clients: Fujitsu, LG, Nespresso, Collins Aerospace, Coca-Cola, PepsiCo.
- Designed and embedded predictive and prescriptive models into executive decision-making workflows, enabling proactive scenario-based planning.
- Built enterprise-grade ETL pipelines in Python, SQL, dbt, Spark, and cloud data warehouse patterns, standardising CRM and e-commerce data across regions into a unified source of truth.
- Created executive dashboards in Power BI and Tableau providing board-level visibility into customer behaviour and revenue growth drivers.

United Nations — IAEA International Atomic Energy Agency | Information Systems Analyst (Internship) | (1 year)

Jul 2020 – Jun 2021 | Vienna, Austria

- Developed microservices pipelines for MTHR analytics with automated experiment tracking; conducted statistical modelling and anomaly detection to support UN staff planning, workforce policy evaluation, and organisational effectiveness.
- Created predictive workforce dashboards in Power BI and SAP BO modelling trends from HRIS and ERP data; trained IAEA staff on applied statistics and data visualisation for mission-support decision-making.

The Hellenic Institute of Cultural Diplomacy | Statistician & Data Analyst | (1 year)

Jul 2019 – Jun 2020 | Athens, Greece

- Applied econometric modelling, hypothesis testing, time series analysis, optimization/linear programming, Monte Carlo simulation, predictive analysis, and statistical reporting for financial datasets supporting international project evaluation and policy strategy.
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>> AI ENGINEERING PROJECTS

BetterPrompt — LLM-Powered Browser Extension |

Jul 2025 – Feb 2026

I engineered for personal reasons [Better Prompt](#), a production Chrome extension and SaaS backend that acts as an AI prompt optimization layer for browser-based AI workflows. The product transforms rough user input into structured, higher-quality prompts through intent analysis, request classification, tone controls, guided refinement, and platform-aware prompt insertion.

Rather than building a simple text rewriter, I designed Better Prompt as a full AI-assisted productivity system: users can draft or select text, optimize it through a side panel or in-page button, refine the result with clarifying questions, copy or edit the improved prompt, and send it into supported AI tools with minimal friction.

Key achievements:

- Built end-to-end a production Chrome Manifest V3 extension and AI SaaS backend that transforms rough user input into structured, optimized prompts across browser-based AI workflows.
- Designed a prompt transformation and LLM orchestration pipeline with intent analysis, Academic / Technical / Creative / General classification, embeddings/RAG-ready retrieval hooks, tone controls, RTF-style prompt scaffolding, guided refinement questions, response validation, caching, and fallback behavior.
- Engineered cross-platform browser integration for AI tools including ChatGPT, Claude, Gemini, Perplexity, Grok, Copilot, Mistral, and DeepSeek using content scripts, DOM adapters, background-worker proxying, side panel UX, and prompt insertion flows.
- Built the production backend with Node.js, Express, Gemini, Anthropic provider routing for creative prompts, Google OAuth verification, Neon/PostgreSQL usage metering, subscription-aware quota enforcement, Revolut payments, signed webhooks, rate limiting, and Railway deployment.
- Developed product UX for templates, tone presets, prompt history, editable optimized output, diff view, guided refinement, upgrade prompts, account settings, and usage status.
- Implemented monetization infrastructure with subscription tiers, daily prompt limits, atomic quota consumption, refund-on-failure handling, coupon logic, payment verification, and webhook replay protection.
- Improved production readiness through API validation, CORS controls, request size limits, timeout-aware background messaging, local fallback optimization, graceful shutdown handling, and Chrome Web Store compliance work.

Mathimatikos.xyz — Production Generative AI Tutoring Platform |

Jan 2025 – July 2026

I engineered the core AI and full-stack infrastructure for [Mathimatikos.xyz](#), a production-grade multilingual math learning platform that transforms text, images, camera captures, and handwriting into structured step-by-step solutions. My work combined AI engineering, backend architecture, frontend systems, database design, payments, authentication, and production reliability.

I designed and implemented multimodal AI workflows for OCR, math extraction, problem solving, LaTeX normalization, structured response parsing, streaming answer generation, graph rendering, and teacher-facing worksheet generation. I built the system around typed API contracts, queue-backed processing, subscription-aware access control, usage metering, and scalable service boundaries rather than a simple chatbot interface.

Key achievements:

- Built multimodal AI pipelines for text input, image OCR, camera capture, and handwriting recognition.
- Designed LLM orchestration flows with RAG (Retrieval-Augmented Generation) patterns, embeddings/vector database architecture, math extraction, step-by-step reasoning, structured JSON outputs, and educational explanations.
- Implemented prompt engineering, response validation, LaTeX normalization, and math-rendering pipelines using KaTeX/MathJax-compatible output.
- Developed streaming AI responses with real-time frontend rendering for long-running solver workflows.
- Created backend services with Express, TypeScript, PostgreSQL/Drizzle, Redis, BullMQ-style queues, and server-side entitlement checks.
- Integrated authentication, subscription tiers, payment lifecycle handling, usage limits, PDF export permissions, and teacher-only generation features.
- Built React/Vite frontend workflows for solver UX, OCR review, generated worksheets, graph output, account settings, pricing, and admin operations.
- Improved production readiness through typed schemas, validation, unit/integration testing, test coverage discipline, rate limiting, error handling, caching, queue fallbacks, SEO, accessibility, and performance optimization.
- Supported SaaS operations with admin dashboards, analytics, coupon management, billing events, feedback handling, and cost-aware AI usage tracking.

AI Early Warning Systems for Disaster Risk Reduction (DRR) |

Oct 2024 – Jun 2025

- International-award-winning suite of deep learning and machine learning models predicting tropical storms, droughts, and floods. Presented at the [UN General Assembly Global Platform](#) for DRR (GP2025) Geneva and featured in the [official virtual exhibition](#).
- Models incorporate feature engineering, transfer learning, predictive modelling, model evaluation, and real-world cloud/container deployment strategies for disaster preparedness and mitigation.